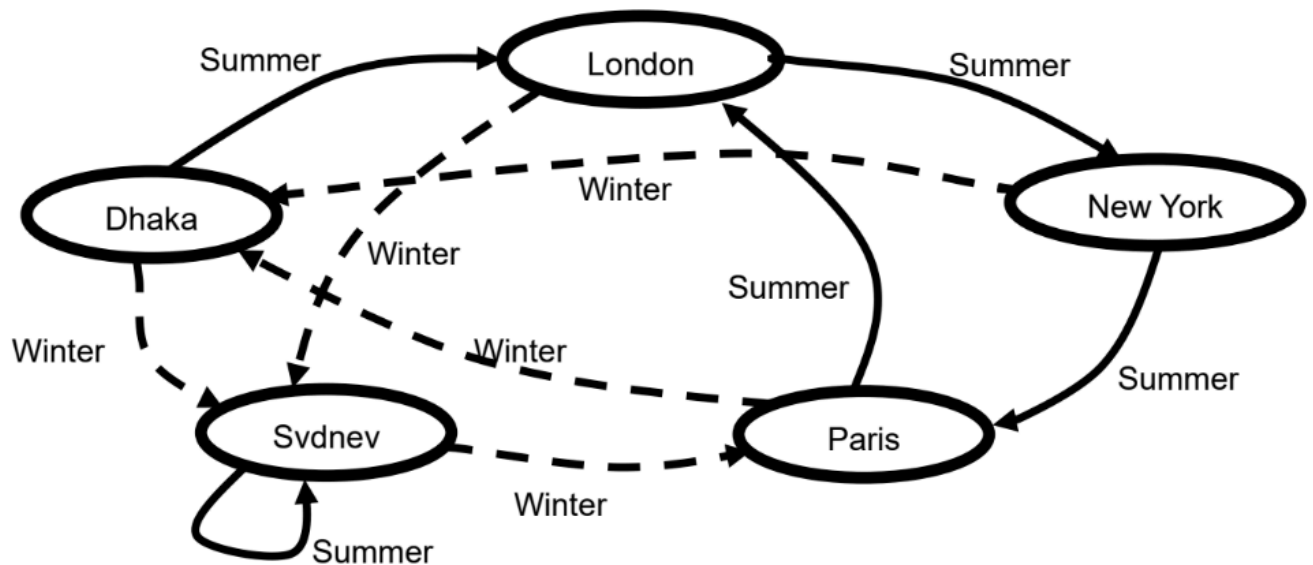


## CSE260: Digital Logic Design - Quiz 4

Time: 25 minutes | Total Mark: 15

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Section: \_\_\_\_\_

1. Dr. Meow Meow, a famous Meow Engineer, travels various cities around the world as per the following diagram. His next destination always depends on his present city as well as the season (Summer or Winter).



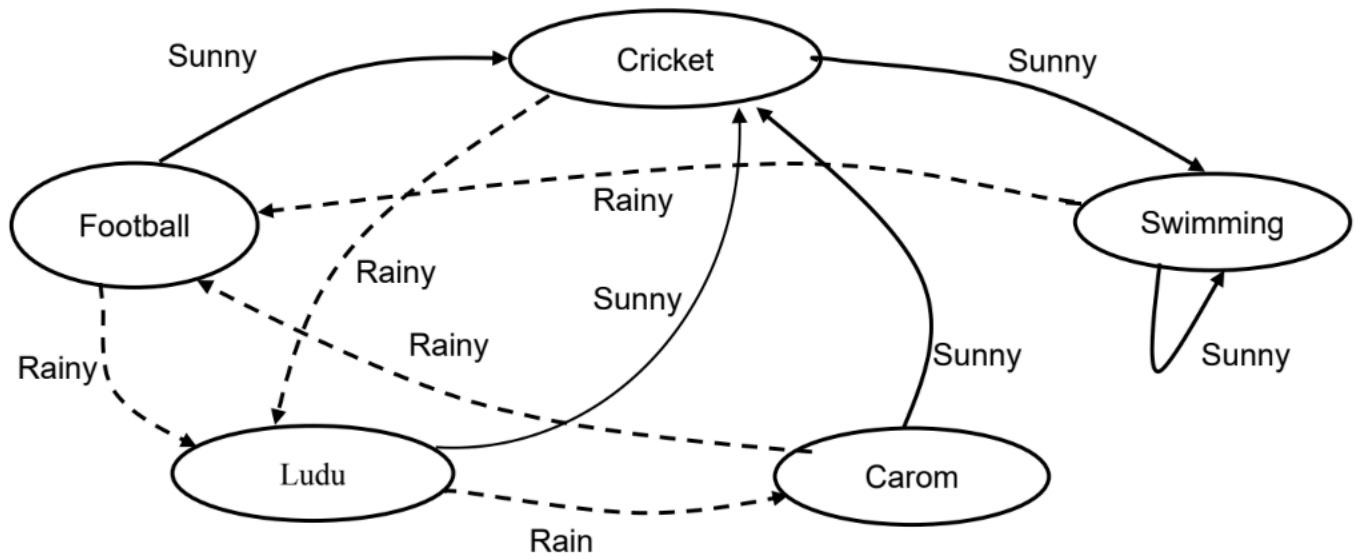
You are asked to design a Synchronous Sequential circuit (with T Flipflops) that will take Season as an input and generate the next destination for Dr. Meow Meow. Show all the necessary steps to generate the circuit. [15]

## CSE260: Digital Logic Design - Quiz 4

Time: 20 minutes | Total Mark: 15

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Section: \_\_\_\_\_

1. Dr. Meow Meow, a famous Electrical Engineering Professor participates in various sports depending on the weather condition (sunny or rainy) as shown in the following diagram.



You are required to design a Synchronous Sequential circuit (Using T flipflop) that will take Weather as an input and generate the next game Dr. Meow Meow will play as shown in the diagram. Show all the necessary steps to generate the circuit. [15]

Date: 11 September 2025

B

CSE260: Digital Logic Design - Quiz 4

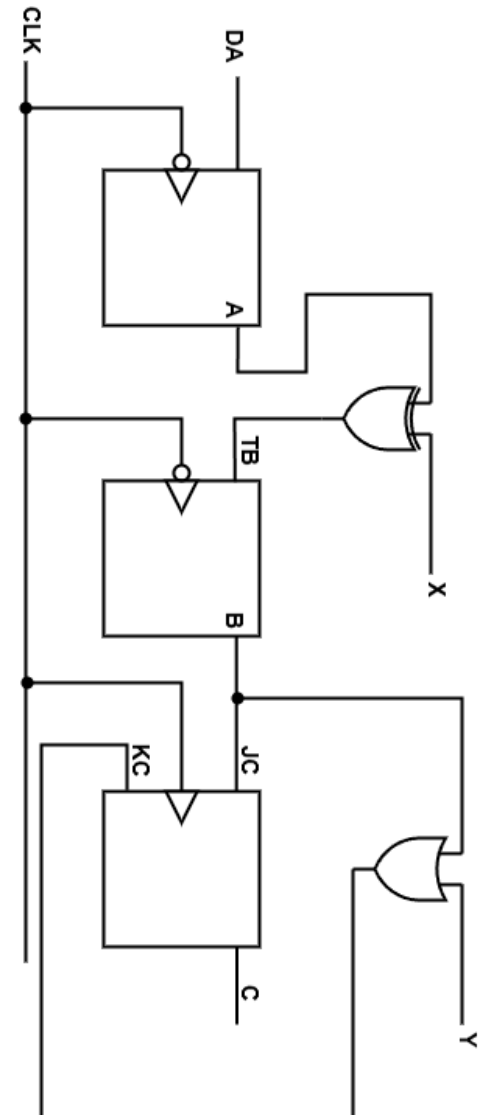
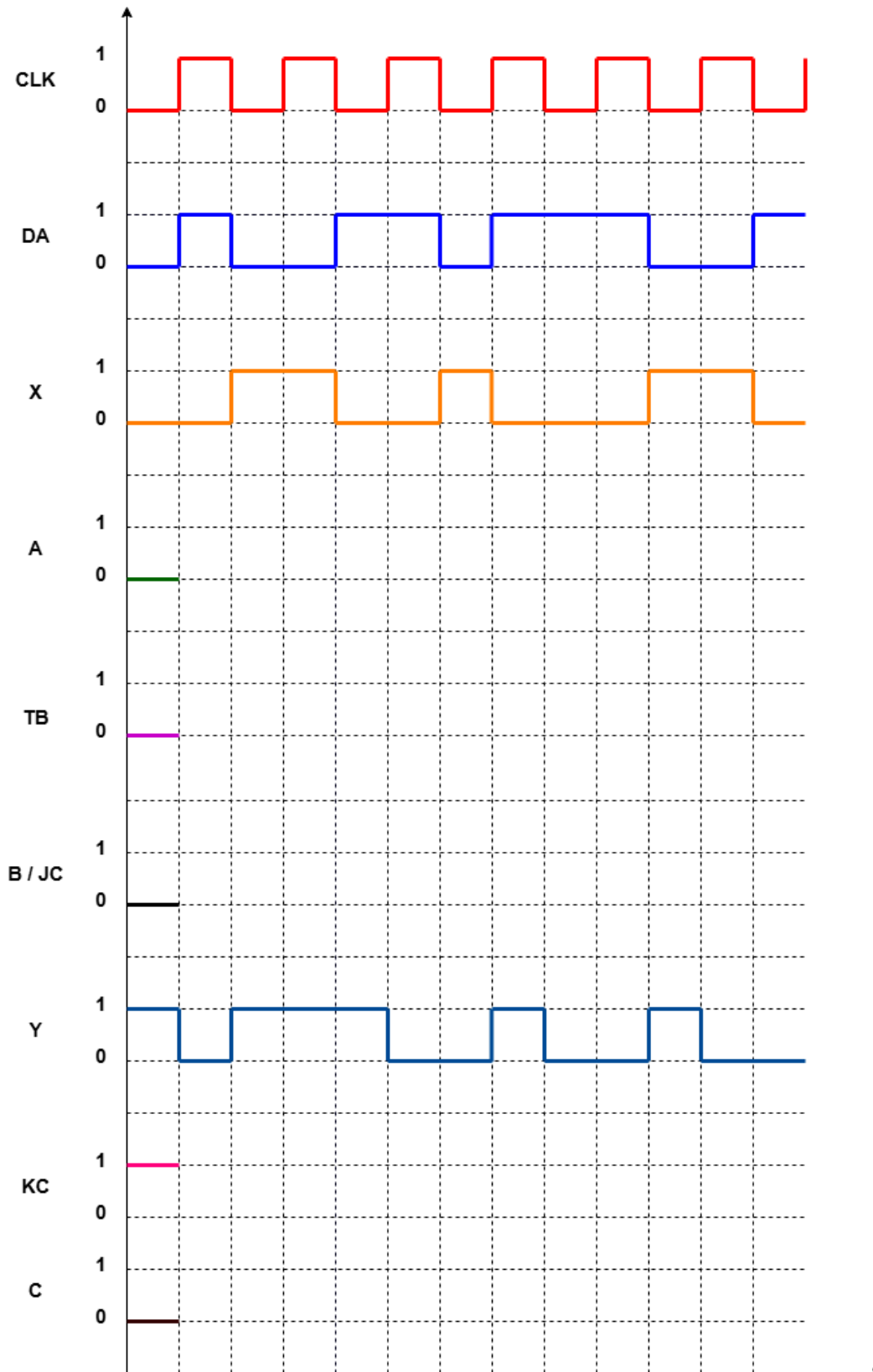
Time: 35 minutes

Total Mark: 15

Name: \_\_\_\_\_ ID: \_\_\_\_\_ Section: \_\_\_\_\_

1. Draw and complete the timing diagram for the following circuit. [8]

Note: The circled clock inputs are negative edge triggered circuits whereas the direct connections are positive edge triggered.



Date: 11 September 2025

2. Design a circuit that will take two 3 bit numbers and do the following operations: [7]

Inputs:  $A = A_2A_1A_0$ ,  $B = B_2B_1B_0$

Output:  $(A+B)-5$  if  $X = 0$  else,  $(A-B)+5$  if  $X = 1$

Equipments allowed: 4 Bit PA, Exclusive Gates, Universal Gates and Not Gates

**Hint: X is the Cin of the Parallel Adders. You need multiple PA to implement the circuit.**

**Show all the +5v and Ground connection where required.**